

Migrating to Claude 4.5

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This guide covers two key migration paths to Claude 4.5 models:

- **Claude Sonnet 3.7 → Claude Sonnet 4.5:** Our most intelligent model with best-in-class reasoning, coding, and long-running agent capabilities
- **Claude Haiku 3.5 → Claude Haiku 4.5:** Our fastest and most intelligent Haiku model with near-frontier performance for real-time applications and high-volume intelligent processing

Both migrations involve breaking changes that require updates to your implementation. This guide will walk you through each migration path with step-by-step instructions and clearly marked breaking changes.

Before starting your migration, we recommend reviewing [What's new in Claude 4.5](#) to understand the new features and capabilities available in these models, including extended thinking, context awareness, and behavioral improvements.

Migrating from Claude Sonnet 3.7 to Claude Sonnet 4.5

Claude Sonnet 4.5 is our most intelligent model, offering best-in-class performance for reasoning, coding, and long-running autonomous agents. This migration includes several breaking changes that require updates to your implementation.

Migration steps

1. Update your model name:

```
# Before (Claude Sonnet 3.7)
model="claude-3-7-sonnet-20250219"

# After (Claude Sonnet 4.5)
model="claude-sonnet-4-5-20250929"
```

2. Update sampling parameters

[Ask Docs](#)

Use only `temperature` OR `top_p` , not both:

```
# Before (Claude Sonnet 3.7) - This will error in Sonnet 4.5
response = client.messages.create(
    model="claude-3-7-sonnet-20250219",
    temperature=0.7,
    top_p=0.9, # Cannot use both
    ...
)

# After (Claude Sonnet 4.5)
response = client.messages.create(
    model="claude-sonnet-4-5-20250929",
    temperature=0.7, # Use temperature OR top_p, not both
    ...
)
```

3. Handle the new `refusal` stop reason

Update your application to handle refusal stop reasons:

```
response = client.messages.create(...)

if response.stop_reason == "refusal":
    # Handle refusal appropriately
    pass
```

4. Update text editor tool (if applicable)

 This is a breaking change from the Claude Sonnet 3.7.

Update to `text_editor_20250728` (type) and `str_replace_based_edit_tool` (name). Remove any code using the `undo_edit` command.

```
# Before (Claude Sonnet 3.7)
tools=[{"type": "text_editor_20250124", "name": "str_replace_editor"}]

# After (Claude Sonnet 4.5)
tools=[{"type": "text_editor_20250728", "name": "str_replace_based_edit_tool"}]
```

Upgrade to `code_execution_20250825`. The legacy version `code_execution_20250522` still works but is not recommended. See [Code execution tool documentation](#) for migration instructions.

6. Remove token-efficient tool use beta header

[Token-efficient tool use](#) is a beta feature that only works with Claude 3.7 Sonnet. All Claude 4 models have built-in token-efficient tool use, so you should no longer include the beta header.

Remove the `token-efficient-tools-2025-02-19` [beta header](#) from your requests:

```
# Before (Claude Sonnet 3.7)
client.messages.create(
    model="claude-3-7-sonnet-20250219",
    betas=["token-efficient-tools-2025-02-19"], # Remove this
    ...
)

# After (Claude Sonnet 4.5)
client.messages.create(
    model="claude-sonnet-4-5-20250929",
    # No token-efficient-tools beta header
    ...
)
```

7. Remove extended output beta header

The `output-128k-2025-02-19` [beta header](#) for extended output is only available in Claude Sonnet 3.7.

Remove this header from your requests:

```
model="claude-3-7-sonnet-20250219",
betas=["output-128k-2025-02-19"], # Remove this
...
)

# After (Claude Sonnet 4.5)
client.messages.create(
    model="claude-sonnet-4-5-20250929",
    # No output-128k beta header
    ...
)
```

8. Update your prompts for behavioral changes

Claude Sonnet 4.5 has a more concise, direct communication style and requires explicit direction. Review [Claude 4 prompt engineering best practices](#) for optimization guidance.

9. Consider enabling extended thinking for complex tasks

Enable [extended thinking](#) for significant performance improvements on coding and reasoning tasks (disabled by default):

```
response = client.messages.create(
    model="claude-sonnet-4-5-20250929",
    max_tokens=16000,
    thinking={"type": "enabled", "budget_tokens": 10000},
    messages=[...]
)
```

 Extended thinking impacts [prompt caching efficiency](#).

10. Test your implementation

Test in a development environment before deploying to production to ensure all breaking changes are properly handled.

Sonnet 3.7 → 4.5 migration checklist

- ☐ Update model ID to `claude-sonnet-4-5-20250929`
- ☐ **BREAKING:** Update sampling parameters to use only `temperature` OR `top_p`, not both
- ☐ Handle new `refusal` stop reason in your application

- ☐ **BREAKING:** Remove any code using the `undo_edit` command (if applicable)
- ☐ Update code execution tool to `code_execution_20250825` (if applicable)
- ☐ Remove `token-efficient-tools-2025-02-19` beta header (if applicable)
- ☐ Remove `output-128k-2025-02-19` beta header (if applicable)
- ☐ Review and update prompts following [Claude 4 best practices](#)
- ☐ Consider enabling extended thinking for complex reasoning tasks
- ☐ Handle `model_context_window_exceeded` stop reason (Sonnet 4.5 specific)
- ☐ Consider enabling memory tool for long-running agents (beta)
- ☐ Consider using automatic tool call clearing for context editing (beta)
- ☐ Test in development environment before production deployment

Features removed from Claude Sonnet 3.7

- **Token-efficient tool use:** The `token-efficient-tools-2025-02-19` beta header only works with Claude 3.7 Sonnet and is not supported in Claude 4 models (see step 6)
- **Extended output:** The `output-128k-2025-02-19` beta header is not supported (see step 7)

Both headers can be included in Claude 4 requests but will have no effect.

Migrating from Claude Haiku 3.5 to Claude Haiku 4.5

Claude Haiku 4.5 is our fastest and most intelligent Haiku model with near-frontier performance, delivering premium model quality with real-time performance for interactive applications and high-volume intelligent processing. This migration includes several breaking changes that require updates to your implementation.

For a complete overview of new capabilities, see [What's new in Claude 4.5](#).

 Haiku 4.5 pricing \$1 per million input tokens, \$5 per million output tokens. See [Claude pricing](#) for details.

Migration steps

1. **Update your model name:**

```
# After (Haiku 4.5)
model="claude-haiku-4-5-20251001"
```

2. Update tool versions (if applicable)

 This is a breaking change from the Claude Haiku 3.5.

Haiku 4.5 only supports the latest tool versions:

```
# Before (Haiku 3.5)
tools=[{"type": "text_editor_20250124", "name": "str_replace_editor"}]

# After (Haiku 4.5)
tools=[{"type": "text_editor_20250728", "name": "str_replace_based_edit_tool"}]
```

- **Text editor:** Use `text_editor_20250728` and `str_replace_based_edit_tool`
- **Code execution:** Use `code_execution_20250825`
- Remove any code using the `undo_edit` command

3. Update sampling parameters

 This is a breaking change from the Claude Haiku 3.5.

Use only `temperature` OR `top_p` , not both:

```
model="claude-3-5-haiku-20241022",
temperature=0.7,
top_p=0.9, # Cannot use both
...
)

# After (Haiku 4.5)
response = client.messages.create(
    model="claude-haiku-4-5-20251001",
    temperature=0.7, # Use temperature OR top_p, not both
    ...
)
```

4. Review new rate limits

Haiku 4.5 has separate rate limits from Haiku 3.5. See [Rate limits documentation](#) for details.

5. Handle the new `refusal` stop reason

Update your application to [handle refusal stop reasons](#).

6. Consider enabling extended thinking for complex tasks

Enable [extended thinking](#) for significant performance improvements on coding and reasoning tasks (disabled by default):

```
response = client.messages.create(
    model="claude-haiku-4-5-20251001",
    max_tokens=16000,
    thinking={"type": "enabled", "budget_tokens": 5000},
    messages=[...]
)
```

 Extended thinking impacts [prompt caching efficiency](#).

7. Explore new capabilities

See [What's new in Claude 4.5](#) for details on context awareness, increased output capacity (64K tokens), higher intelligence, and improved speed.

8. Test your implementation

Test in a development environment before deploying to production to ensure all breaking changes are properly handled.

- ☐ Update model ID to `claude-haiku-4-5-20251001`
- ☐ **BREAKING:** Update tool versions to latest (e.g., `text_editor_20250728` , `code_execution_20250825`) - legacy versions not supported
- ☐ **BREAKING:** Remove any code using the `undo_edit` command (if applicable)
- ☐ **BREAKING:** Update sampling parameters to use only `temperature` OR `top_p` , not both
- ☐ Review and adjust for new rate limits (separate from Haiku 3.5)
- ☐ Handle new `refusal` stop reason in your application
- ☐ Consider enabling extended thinking for complex reasoning tasks (new capability)
- ☐ Leverage context awareness for better token management in long sessions
- ☐ Prepare for larger responses (max output increased from 8K to 64K tokens)
- ☐ Review and update prompts following [Claude 4 best practices](#)
- ☐ Test in development environment before production deployment

Choosing between Sonnet 4.5 and Haiku 4.5

Both Claude Sonnet 4.5 and Claude Haiku 4.5 are powerful Claude 4 models with different strengths:

Choose Claude Sonnet 4.5 (most intelligent) for:

- **Complex reasoning and analysis:** Best-in-class intelligence for sophisticated tasks
- **Long-running autonomous agents:** Superior performance for agents working independently for extended periods
- **Advanced coding tasks:** Our strongest coding model with advanced planning and security engineering
- **Large context workflows:** Enhanced context management with memory tool and context editing capabilities
- **Tasks requiring maximum capability:** When intelligence and accuracy are the top priorities

Choose Claude Haiku 4.5 (fastest and most intelligent Haiku) for:

- **Real-time applications:** Fast response times for interactive user experiences with near-frontier performance
- **High-volume intelligent processing:** Cost-effective intelligence at scale with improved speed
- **Cost-sensitive deployments:** Near-frontier performance at lower price points
- **Sub-agent architectures:** Fast, intelligent agents for multi-agent systems

Extended thinking recommendations

Claude 4 models, particularly Sonnet and Haiku 4.5, show significant performance improvements when using extended thinking for coding and complex reasoning tasks. Extended thinking is **disabled by default** but we recommend enabling it for demanding work.

Important: Extended thinking impacts prompt caching efficiency. When non-tool-result content is added to a conversation, thinking blocks are stripped from cache, which can increase costs in multi-turn conversations. We recommend enabling thinking when the performance benefits outweigh the caching trade-off.

Other migration scenarios

The primary migration paths covered above (Sonnet 3.7 → 4.5 and Haiku 3.5 → 4.5) represent the most common upgrades. However, you may be migrating from other Claude models to Claude 4.5. This section covers those scenarios.

Migrating from Claude Sonnet 4 → Sonnet 4.5

Breaking change: Cannot specify both `temperature` and `top_p` in the same request.

All other API calls will work without modification. Update your model ID and adjust sampling parameters if needed:

```
# Before (Claude Sonnet 4)
model="claude-sonnet-4-20250514"

# After (Claude Sonnet 4.5)
model="claude-sonnet-4-5-20250929"
```

Migrating from Claude Opus 4.1 → Sonnet 4.5

No breaking changes. All API calls will work without modification.

Simply update your model ID:

```
# After (Claude Sonnet 4.5)
model="claude-sonnet-4-5-20250929"
```

Claude Sonnet 4.5 is our most intelligent model with best-in-class reasoning, coding, and long-running agent capabilities. It offers superior performance compared to Opus 4.1 for most use cases.

Migrating from Claude Opus 4.1 → Opus 4.5

No breaking changes. All API calls will work without modification.

Simply update your model ID:

```
# Before (Claude Opus 4.1)
model="claude-opus-4-1-20250805"

# After (Claude Opus 4.5)
model="claude-opus-4-5-20251101"
```

Claude Opus 4.5 is our most intelligent model, combining maximum capability with practical performance. It features step-change improvements in vision, coding, and computer use at a more accessible price point than Opus 4.1. Ideal for complex specialized tasks and professional software engineering.

 For codebases with many model references, a [Claude Code plugin](#) is available to automate migration to Opus 4.5.

Migrating between Claude 4.5 models

No breaking changes. All API calls will work without modification.

Simply update your model ID.

Need help?

- Check our [API documentation](#) for detailed specifications
- Review [model capabilities](#) for performance comparisons
- Review [API release notes](#) for API updates



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